

SIDNEY, NE, USA

WMO: 725610

Lat: 41.100N

Lon: 102.983W

Elev: 1312

StdP: 86.52

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24030

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-19.6	-16.5	-23.1	0.5	-19.0	-19.9	0.7	-14.9	16.9	-1.0	15.2	-2.2	4.3	280	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	36.2	17.9	34.4	17.7	32.7	17.6	20.6	29.3	19.9	28.9	19.1	28.2	5.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	15.3	22.5	17.4	14.6	22.1	16.5	13.8	22.1	65.1	29.0	62.5	28.7	60.0	28.1	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.2	10.8	-25.0	38.7	4.1	1.6	-28.0	39.9	-30.4	40.8	-32.7	41.7	-35.7	42.9		
(5)			-25.0	21.9	4.1	0.9	-28.0	22.5	-30.4	23.1	-32.7	23.5	-35.7	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-1.9	-1.3	3.8	7.6	13.1	19.6	23.6	22.2	17.4	9.3	2.9	-2.1
	DBStd	10.48	6.08	6.47	5.90	5.12	4.82	4.07	2.96	3.09	4.69	5.31	5.71	6.19
	HDD10.0	1694	370	318	203	103	26	1	0	0	5	76	217	376
	HDD18.3	3618	628	551	452	323	173	33	3	7	71	281	462	634
	CDD10.0	1536	0	1	10	31	121	288	422	378	227	54	5	0
	CDD18.3	418	0	0	0	0	9	71	166	127	43	1	0	0
	CDH23.3	5971	0	0	10	44	242	1037	2149	1620	758	107	4	0
	CDH26.7	2894	0	0	0	4	74	474	1168	807	338	28	0	0
(14) Wind	WSAvg	5.3	5.4	5.6	5.8	6.0	5.3	4.9	4.6	4.6	4.8	5.3	5.3	5.5
(15) Precipitation	PrecAvg	436	6	8	19	47	64	80	66	54	39	31	12	7
	PrecMax	618	17	26	55	110	191	209	124	112	144	114	30	27
	PrecMin	192	0	0	2	5	6	21	29	6	4	1	0	0
	PrecStd	94	5	5	15	26	39	45	25	25	30	23	8	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	19.9	24.9	27.3	32.0	37.6	38.5	37.2	35.4	29.7	23.5	17.0
		MCWB	6.4	7.6	10.2	12.0	15.4	17.2	18.0	17.5	16.9	13.5	10.2	6.7
	2%	DB	13.5	15.6	21.3	24.3	28.7	34.2	36.7	35.1	32.8	26.7	19.7	13.2
		MCWB	5.0	5.9	8.9	11.2	15.1	17.2	18.3	17.6	16.0	12.7	8.6	4.9
	5%	DB	10.3	12.1	18.0	21.5	26.1	31.9	35.0	33.3	30.9	23.0	16.6	10.2
		MCWB	3.5	4.4	7.6	10.4	14.5	17.2	18.1	17.7	15.7	11.3	7.2	3.4
	10%	DB	7.3	8.6	14.8	18.4	23.2	29.2	33.0	31.5	28.1	19.8	12.8	7.3
		MCWB	2.0	2.6	6.2	9.3	13.4	17.0	18.3	17.7	15.1	10.2	5.5	1.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	8.0	11.3	13.9	18.0	21.0	21.7	21.2	19.3	15.1	10.9	7.1
		MCDB	15.3	18.5	21.7	23.6	26.4	28.9	29.9	29.7	29.1	25.3	21.5	16.0
	2%	WB	5.2	6.2	9.5	12.1	16.6	19.8	20.8	20.2	18.0	13.6	9.0	5.3
		MCDB	13.0	14.8	19.4	21.1	24.6	28.0	29.6	29.0	27.3	23.1	18.9	12.9
	5%	WB	3.7	4.6	8.0	10.9	15.4	18.7	20.1	19.5	17.1	12.2	7.5	3.6
		MCDB	9.8	11.5	17.1	20.4	23.3	27.9	29.3	28.3	26.3	21.0	16.1	9.6
	10%	WB	2.2	2.9	6.7	9.8	14.2	17.8	19.5	18.8	16.1	10.8	5.7	2.0
		MCDB	6.9	8.1	14.3	17.6	22.0	26.8	28.7	27.5	25.1	18.6	12.4	7.1
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	13.9	15.7	15.0	14.7	16.0	16.5	16.3	16.6	15.6	15.1	13.8
		MCDBR	18.1	19.3	21.4	21.3	19.7	20.0	19.9	19.6	20.9	21.8	20.7	18.0
		MCWBR	11.0	11.2	11.3	10.4	8.6	7.0	6.0	6.2	7.6	10.3	11.5	11.0
	5% WB	MCDBR	17.0	18.3	20.0	19.3	16.9	17.2	16.2	16.1	17.4	18.8	19.9	17.1
		MCWBR	10.7	10.8	10.9	10.0	8.0	7.3	5.9	6.1	7.1	9.4	11.2	10.7
(40) Clear-Sky Solar Irradiance	taub		0.246	0.253	0.276	0.301	0.325	0.336	0.364	0.355	0.316	0.279	0.257	0.247
	taud		2.586	2.565	2.521	2.457	2.459	2.455	2.412	2.425	2.511	2.588	2.570	2.559
	Ebn at Noon		944	982	985	972	949	935	905	905	925	936	917	911
	Edh at Noon		66	79	93	107	110	111	114	110	94	77	67	63
(44) All-Sky Solar Radiation	RadAvg		2.39	3.21	4.46	5.47	6.32	7.37	7.25	6.35	5.16	3.61	2.58	2.02
	RadStd		0.12	0.19	0.33	0.34	0.49	0.39	0.28	0.22	0.25	0.30	0.15	0.12

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page